

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and **COURSE CODE:** CSA1522
Big Data Analytics

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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 3

Code of Conduct

I Kanmani L(192511414) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks - Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

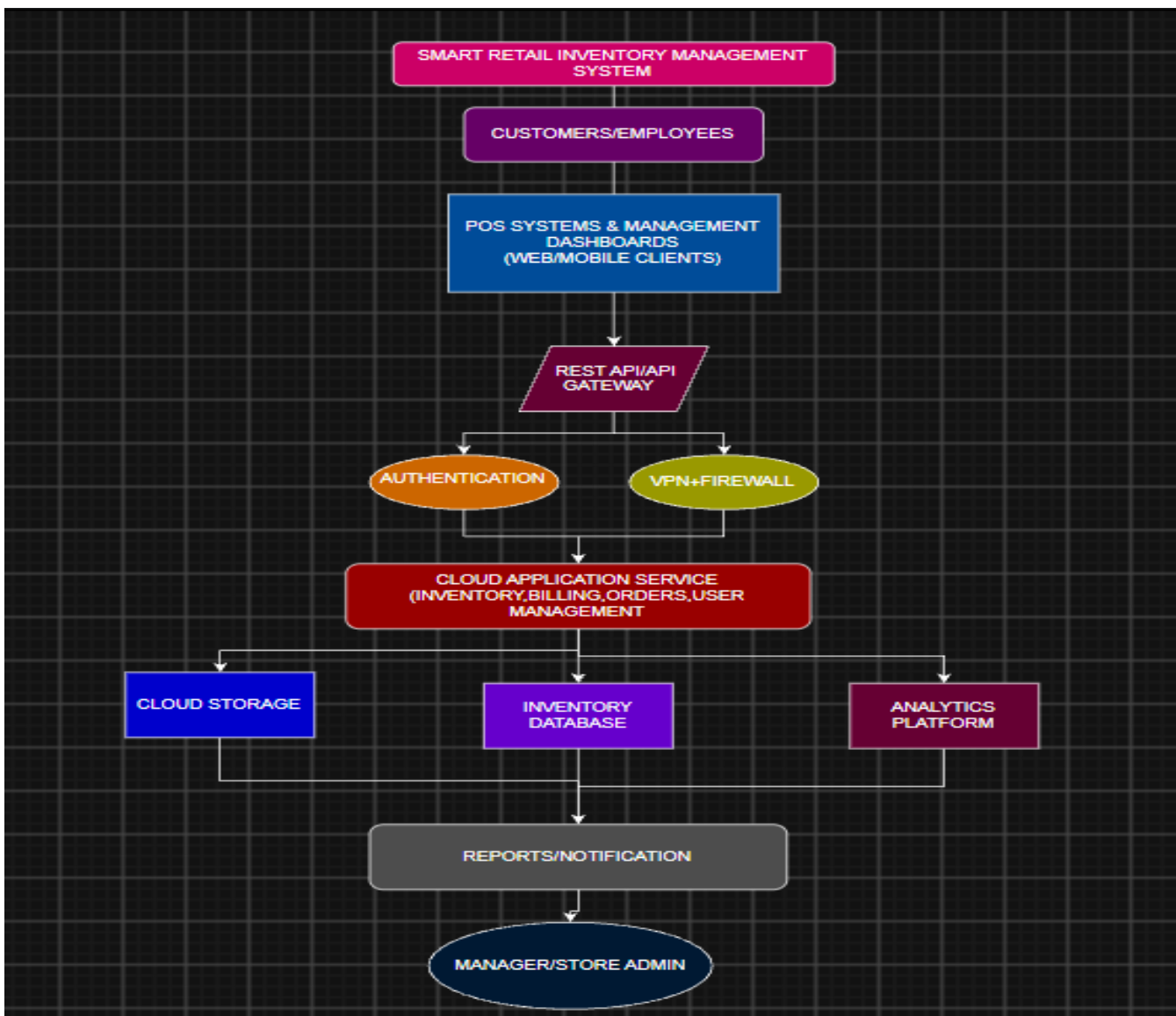
1. Case Study 1: Smart Retail Inventory Management System

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Answer:

A Smart Retail Inventory Management System is a cloud-based application that helps retail businesses manage products, inventory, sales transactions, and customer information in a centralized manner.

The Point of Sale (POS) systems and management dashboards communicate with cloud services through APIs. The cloud stores inventory details, transaction logs, and analytics datasets, while platform services analyze the data to predict future demand.



Working of the System

The entire retail chain is connected to a centralized cloud platform. Every store uses a POS system to record customer purchases and inventory updates.

1. When a customer purchases a product, the POS system records the transaction.
2. The transaction details are sent to the cloud using secure APIs.
3. Cloud services receive the request and update the inventory database immediately.
4. The updated stock information becomes available to all branches in real time.
5. Product details, customer transactions, and sales records are stored securely in cloud storage.
6. Platform services analyze the collected data and generate reports.
7. Demand forecasting algorithms study previous sales patterns and estimate future product requirements.
8. Managers use dashboards to monitor stock levels, sales performance, and business reports from any location.
9. The cloud automatically synchronizes information across all stores, ensuring consistency and accuracy.

Cloud Components Used:

➤ Client Layer

The client layer includes POS terminals, manager dashboards, and web applications. These devices allow employees to access inventory information, process sales, and generate reports.

➤ API Layer

APIs act as the communication bridge between client devices and cloud services. They transfer inventory updates, product information, and transaction data securely and quickly.

➤ Cloud Services

Cloud services handle all business logic such as inventory updates, transaction processing, report generation, and user authentication. They ensure that all operations are performed efficiently.

➤ Cloud Storage

Cloud storage stores:

- Product information
- Sales records
- Transaction logs
- Customer purchase history

Platform Services:

Platform services perform:

- Sales analytics
- Demand forecasting
- Business intelligence
- Report generation
- Application management

Cloud Service Models:

1. Infrastructure as a Service (IaaS)

IaaS provides the infrastructure required to run the application.

It includes:

- Virtual machines
- Cloud storage
- Networking
- Load balancers
- Backup services

The retailer does not need to purchase expensive physical servers because the cloud provider manages the infrastructure.

2. Platform as a Service (PaaS)

PaaS provides the platform for developing and managing cloud applications.

It offers:

- API management
- Analytics tools
- Database services
- Demand forecasting
- Development environment

This helps developers build and maintain applications more efficiently.

3. Software as a Service (SaaS)

The inventory management software is delivered as SaaS.

Employees can:

- Access dashboards
- Monitor inventory
- Generate reports
- Track sales
- Manage products

Security Features

VPN (Virtual Private Network)

- Creates a secure connection between retail stores and cloud servers.
- Protects sensitive data while it travels over the internet.

Firewall

- Monitors incoming and outgoing traffic.
- Prevents unauthorized users from accessing the cloud.
- Protects the system from cyber attacks.

API Gateway

- Authenticates every API request.
- Prevents unauthorized access.
- Improves application security.

Access Control and Encryption

- User authentication ensures only authorized employees can access the system.
- Data encryption protects business information during transmission and storage.

Advantages

- Real-time inventory updates.
- Centralized inventory management for all branches.
- Accurate demand forecasting.
- Faster business decisions.
- Reduced hardware and maintenance costs.

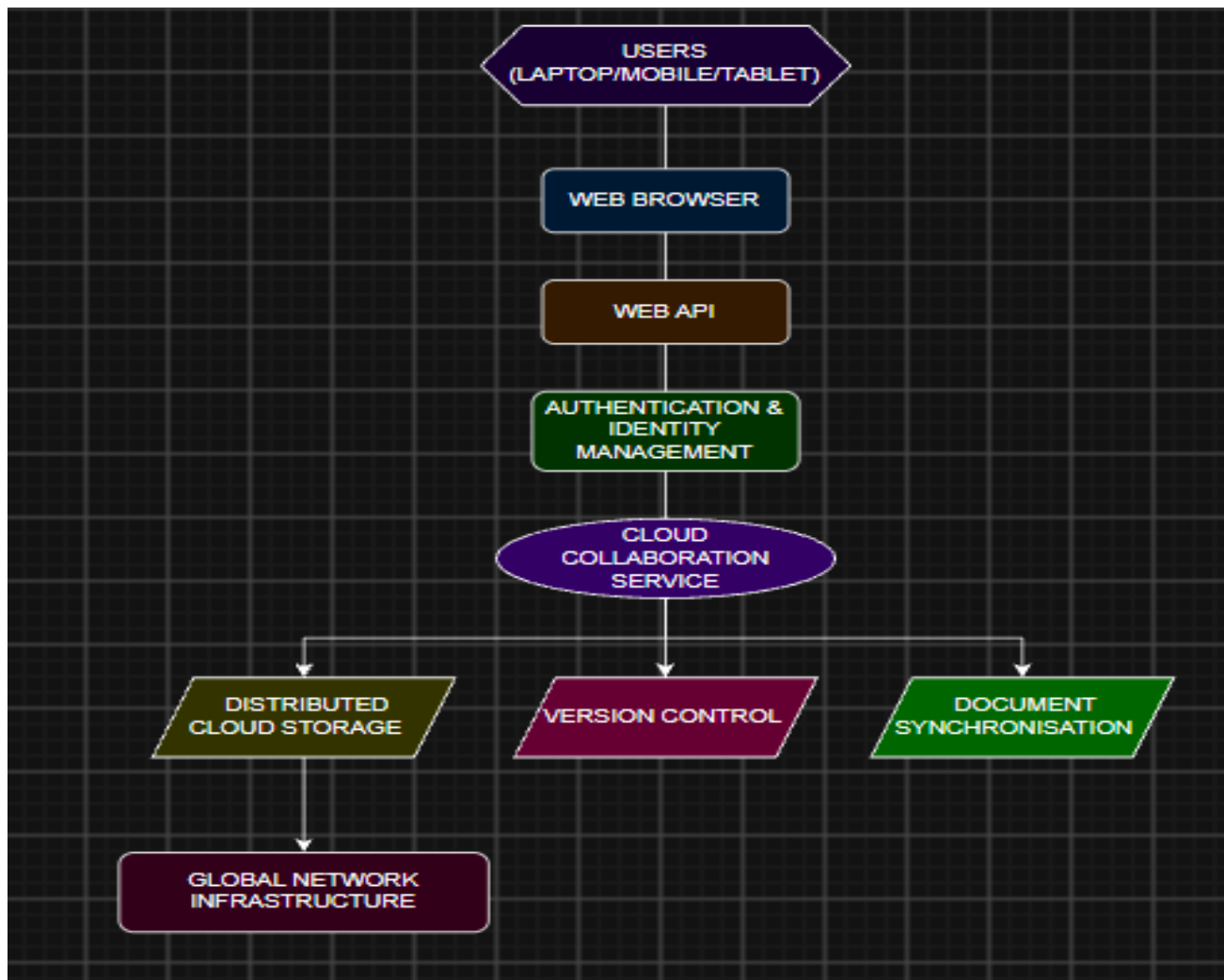
2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

Answer:

A Cloud-Based Document Collaboration System is an online platform that allows multiple users to create, edit, and share documents simultaneously from any location. Applications like Google Docs are examples of such systems. Users can access the platform using a web browser without installing additional software.

The system stores documents in distributed cloud storage, synchronizes changes across devices using web APIs, and provides encryption, identity management, and access control.



Working of the System

1. The document collaboration system allows users to create, edit, and share documents through a web browser using internet.
2. When a user opens or edits a document, the request is sent to the cloud through web APIs.
3. The cloud collaboration service processes the request and updates the document instantly
4. All documents are stored in distributed cloud storage instead of a single server. If one server fails, another server provides access to the same document, ensuring continuous availability.
5. The system also maintains version control, which records every change made to a document.
6. Users can view previous versions, restore earlier copies if required.
7. The global network infrastructure ensures low latency by serving users from the nearest available data center. It also provides high availability so that the service remains accessible even if one server or region experiences a failure.

Cloud Components Used

Client Layer

The client layer includes:

- Web browsers
- Mobile devices
- Tablets
- Desktop and laptop computers

Web APIs

Web APIs enable communication between users' devices and cloud services.

They are responsible for:

- Sending document updates
- Managing user requests
- Supporting real-time collaboration

Cloud Collaboration Services

Cloud services manage:

- Document editing
- Real-time collaboration
- User authentication
- File sharing

Distributed Cloud Storage

Distributed storage stores documents across multiple cloud servers.

It provides:

- High availability
- Automatic backup
- Data redundancy
- Faster access
- Better reliability

Version Control

Version control:

- Saves every modification
- Maintains document history
- Allows rollback to previous versions
- Tracks changes made by different users

Cloud Service Models

Infrastructure as a Service (IaaS)

IaaS provides:

- Virtual servers
- Networking
- Storage infrastructure
- Load balancing
- Backup and recovery

Platform as a Service (PaaS)

PaaS provides:

- Development environment
- Database services
- Web API management
- Synchronization services
- Application hosting

Software as a Service (SaaS)

The collaboration application itself is delivered as SaaS.

Users can:

- Create documents
- Edit files
- Share documents
- Collaborate in real time
- Access files from anywhere

Security Features

The collaboration platform uses several security mechanisms to protect user data.

Encryption

- Encrypts documents during storage and transmission.
- Prevents unauthorized users from reading sensitive information.

Identity Management

- Verifies user identity before granting access.
- Supports secure login using usernames, passwords, or multi-factor authentication.

Access Permissions

- Allows document owners to decide who can view, edit, or comment.
- Prevents unauthorized modifications.

Network Security

- Secure communication protocols protect data exchanged between users and cloud servers.
- Continuous monitoring helps detect unauthorized access and cyber threats.

Advantages

- Enables real-time collaboration from any location.
- Supports multiple users editing the same document simultaneously.
- Documents are automatically saved in the cloud.
- Version control allows recovery of previous document versions.
- Distributed storage improves reliability and availability.
- Low latency provides faster document access.

The Cloud-Based Document Collaboration System is an effective cloud computing solution that enables users to create, edit, and share documents in real time. It combines distributed cloud storage, web APIs, version control, and cloud collaboration services to provide seamless document management across multiple devices.

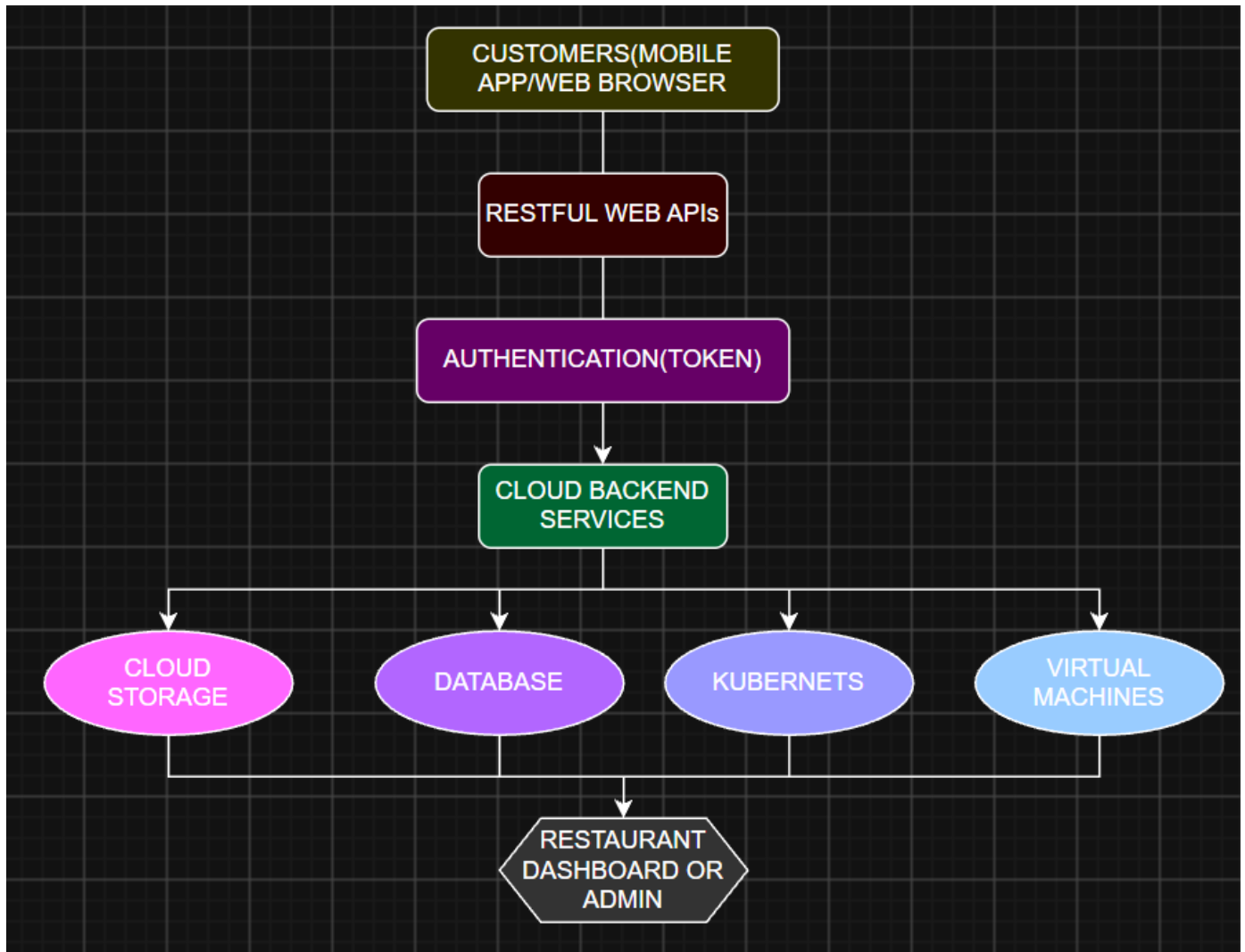
3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Answer:

An Online Food Ordering Platform is a cloud-based application that allows customers to browse restaurant menus, place food orders, make payments, and track deliveries through web browsers or mobile applications.

In this case, the startup uses cloud computing to provide a scalable, reliable, and secure platform.. The application is deployed using virtual machines and managed Kubernetes to handle increasing user traffic efficiently.



Working of the System

- 1.The online food ordering platform allows customers to access the application through a web browser or a mobile application.
- 2.Users can browse restaurant menus, select food items, place orders, make online payments, and track their orders in real time.
- 3.When a customer places an order, the frontend sends the request to the backend through RESTful Web APIs.
- 4.The backend verifies the user's identity, processes the order, updates the database, and sends a confirmation to both the customer and the restaurant.
- 5.Menu images, invoices, customer details, and other important files are stored securely in cloud storage..
- 6.Kubernetes automatically distributes the workload, manages containers, and increases or decreases application instances depending on user demand.

7. Restaurant owners use an admin dashboard to receive orders, update menu items, and monitor business performance.

8. Customers receive instant notifications regarding order confirmation, preparation, and delivery status.

Cloud Components Used

Client Layer

The client layer consists of:

- Web browsers
- Mobile applications
- Restaurant admin dashboard

These clients allow users and restaurant owners to interact with the platform.

RESTful Web APIs

RESTful APIs act as the communication bridge between the frontend and backend.

They are responsible for:

- Sending food orders
- Retrieving menu details
- Processing payments
- Tracking delivery status

Cloud Backend Services

Backend services perform:

- Order processing
- Payment handling
- User authentication
- Restaurant management
- Notification services

Cloud Storage

Cloud storage securely stores:

- Menu images
- Customer details
- Digital invoices

- Order history
- Restaurant information

Virtual Machines

Virtual machines provide:

- Computing resources
- Memory
- Networking
- Application hosting

They support the execution of backend services.

Managed Kubernetes

Managed Kubernetes:

- Deploys application containers
- Automatically scales the application
- Balances workload
- Improves reliability
- Supports high availability during heavy traffic

Cloud Service Models

- Infrastructure as a Service (IaaS)

IaaS provides the infrastructure required to run the application.

It includes:

- Virtual machines
- Networking
- Storage
- Load balancing
- Backup services

These resources ensure reliable application performance.

➤ Platform as a Service (PaaS)

PaaS provides:

- Managed Kubernetes
- API hosting
- Development environment
- Database management
- Monitoring services

It simplifies application deployment and maintenance.

➤ Software as a Service (SaaS)

The food ordering application is delivered as SaaS.

Customers can:

- Browse menus
- Place food orders
- Make payments
- Track deliveries

Restaurant owners can manage menus, orders, and customer requests using the same platform.

Security Features

The platform implements several security mechanisms to protect user and business data.

Authentication Tokens

- Verify the identity of users before allowing access.
- Prevent unauthorized users from accessing the system.
- Secure user sessions after login.

HTTPS

- Encrypts communication between clients and cloud servers.
- Protects sensitive information such as passwords and payment details.
- Prevents data interception during transmission.

Role-Based Access Control (RBAC)

- Provides different access levels for customers, restaurant owners, delivery partners, and administrators.
- Ensures users can access only the resources related to their role.
- Improves overall system security.

Advantages

- Customers can order food anytime using web or mobile applications.
- RESTful APIs enable fast communication between frontend and backend.
- Cloud storage provides secure and centralized data management.
- Virtual machines offer reliable computing resources.
- Managed Kubernetes automatically scales the application during high traffic.
- Secure authentication and HTTPS protect user information.

The Online Food Ordering Platform demonstrates how cloud computing can be used to build a secure, scalable, and reliable application. It integrates clients, RESTful Web APIs, cloud storage, virtual machines, and managed Kubernetes to provide real-time food ordering and delivery services.